SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology / College of Natural and Applied Sciences

Department of Informatics and Information Technology (DIIT)

BIT 111: MICROCOMPUTERS AND APPLICATIONS

TAKEHOME HOMEWORK ASSIGNMENT 5 (TOPIC 5)

Theme: MICROCOMPUTER APPLICATIONS: WORD PROCESSING, SIMPLE SPREADSHEETS & INTERNET PROTOCOLS

Course Instructor: Madam Bertha Msuliche Lebalwa (MSc)
Curriculum Coverage: Subtopic 5.1 (Word Processors & Formatting), Subtopic 5.2 (Spreadsheets & Simple Formulas), Subtopic 5.3 (Web Browsing & Email Protocols)
Assessment Weight: Coursework Component (Official Scheme: Assignments 5%, Tests 15%, Practical Report 10%, Practical Test 30%, UE 40% - subject to instructor review)
Submission Deadline: End of Week 10 by 17:00 EAT | Total Allocation: 100 Marks

INSTRUCTIONS TO CANDIDATES:
1. This assignment must be solved individually. Plagiarism, peer copying, or unreferenced AI generation will result in an automatic score of zero and disciplinary reporting under SUA Examination Regulations.
2. Answer ALL questions in Sections A, B, and C. Allocate your effort according to the mark allocations shown in brackets.
3. Support your explanations with clear examples, labeled diagrams, or command syntax where requested.
4. Use spreadsheet formulas only where requested in the spreadsheet assignment; no other numerical calculations are required.
5. Submit your completed work as a clean, stapled technical report or upload the single consolidated PDF to the SUA Virtual Learning Environment (Moodle) before the deadline.

SECTION A: FOUNDATIONAL SOFTWARE CONCEPTS [25 MARKS]

Question 1 [5 Marks]: In word processing systems (e.g., Microsoft Word, LibreOffice Writer), differentiate between a 'Section Break (Next Page)' and a simple 'Page Break'. Describe a specific scenario in compiling an academic undergraduate thesis where a Section Break is mandatory.

Question 2 [5 Marks]: Explain the mechanical architecture of Mail Merge. Identify the three principal components involved in any mail merge pipeline and explain how conditional fields (`IF...THEN...ELSE`) enable dynamic document customization.

Question 3 [5 Marks]: In a spreadsheet, explain how a cell reference such as `B2` identifies a value by its column and row. Describe why a formula can refer to a cell rather than repeat a value.

Question 4 [5 Marks]: Explain the difference between typing a value directly into a cell and using a simple formula that refers to other cells. Give one classroom example of when a formula is useful.

Question 5 [5 Marks]: Contrast the two primary electronic mail retrieval protocols: POP3 (Post Office Protocol v3) and IMAP4 (Internet Message Access Protocol v4). Under what operational conditions is IMAP4 strictly required for university staff and students?

SECTION B: SIMPLE SPREADSHEET FORMULAS & PROTOCOLS [35 MARKS]

Question 6 [15 Marks]: A farm records daily milk yields in cells B2 through B6. Write spreadsheet formulas to add the yields with `SUM`, find their mean with `AVERAGE`, and calculate the value in a row by multiplying quantity in cell C2 by price in cell D2. Explain what each formula does. [15 Marks]

Question 7 [10 Marks]: A forestry class records seedling counts in one column and planted counts in another. Explain how to enter a simple addition formula and use `SUM` to find a column total. State how copying a formula to another row changes its cell references. [10 Marks]

Question 8 [10 Marks]: Web Protocols & HTTP/HTTPS Security Cycle:
(a) Dissect the components of the Uniform Resource Locator (URL): `https://www.sua.ac.tz:443/academics/courses?dept=diit&term;=2#syllabus`. Identify: Protocol/Scheme, Fully Qualified Domain Name (FQDN), Port, Path, Query String, and Anchor/Fragment. [4 Marks]
(b) Explain the HTTP Request-Response cycle. Differentiate between the HTTP `GET` and `POST` methods regarding parameter transmission and security. [3 Marks]
(c) Interpret the following HTTP response status codes: 200 OK, 301 Moved Permanently, 401 Unauthorized, 404 Not Found, 500 Internal Server Error, and 503 Service Unavailable. [3 Marks]

SECTION C: APPLIED DOCUMENT & ENTERPRISE ARCHITECTURE [40 MARKS]

Question 9 [15 Marks]: Academic Document Engineering & Master Dissertation Compilation:
A postgraduate student at SUA is formatting a 120-page dissertation in Microsoft Word / LibreOffice Writer requiring strict university formatting standards:
- Preliminary pages (Title, Declaration, Abstract, Table of Contents) must use Roman numerals (i, ii, iii, iv) centered at the bottom.
- Chapter 1 onwards must begin with Arabic numeral '1' at the top-right corner.
- Chapter 4 contains a wide 8-column agricultural field trial table that must be displayed in Landscape orientation, while preceding and succeeding pages remain Portrait.
(a) Provide step-by-step technical instructions detailing how to insert Section Breaks, unlink headers and footers ('Link to Previous'), configure page numbering formats, and isolate page orientation for Chapter 4 without corrupting the rest of the document. [8 Marks]
(b) Explain how the automated Table of Contents (TOC) and Table of Figures are constructed using Heading Styles (Heading 1, Heading 2) and Caption tags. Why should a professional academic document never contain manually typed tables of contents? [4 Marks]
(c) Explain the utility of the 'Track Changes' feature and document compare tools in collaborative academic review between students and thesis supervisors. [3 Marks]

Question 10 [15 Marks]: Electronic Mail Infrastructure & Anti-Spoofing Architecture:
The SUA ICT Directorate operates the university mail server domain `@sua.ac.tz`. Recently, fraudulent emails claiming to be from the Vice Chancellor's office were received by students.
(a) Trace the complete path of an email sent via SMTP from `instructor@sua.ac.tz` to `student@gmail.com`. Detail the interactions between Mail User Agent (MUA), Mail Submission Agent (MSA), Mail Transfer Agent (MTA), and Mail Delivery Agent (MDA). [6 Marks]
(b) Explain how the three foundational email authentication protocols protect against domain spoofing and phishing: [6 Marks]
1. SPF (Sender Policy Framework): How DNS TXT records specify authorized sending mail servers.
2. DKIM (DomainKeys Identified Mail): How public-key digital signatures in message headers verify message integrity.
3. DMARC (Domain-based Message Authentication, Reporting, and Conformance): How policy enforcement and reporting align SPF and DKIM results.
(c) Construct a valid DNS TXT record for SPF that authorizes the mail server with IP `196.216.180.15` to send email for `sua.ac.tz` and instructs receiving servers to hard-fail (`-all`) all other sources. [3 Marks]

Question 11 [10 Marks]: A class records seedling types and quantities in a spreadsheet.
(a) Describe how to label columns and enter one record per row. [3 Marks]
(b) Explain how to sort the records by seedling type and make a simple chart to compare categories. [4 Marks]
(c) Write a simple `SUM` formula for adding the quantities in cells B2 through B6. [3 Marks]